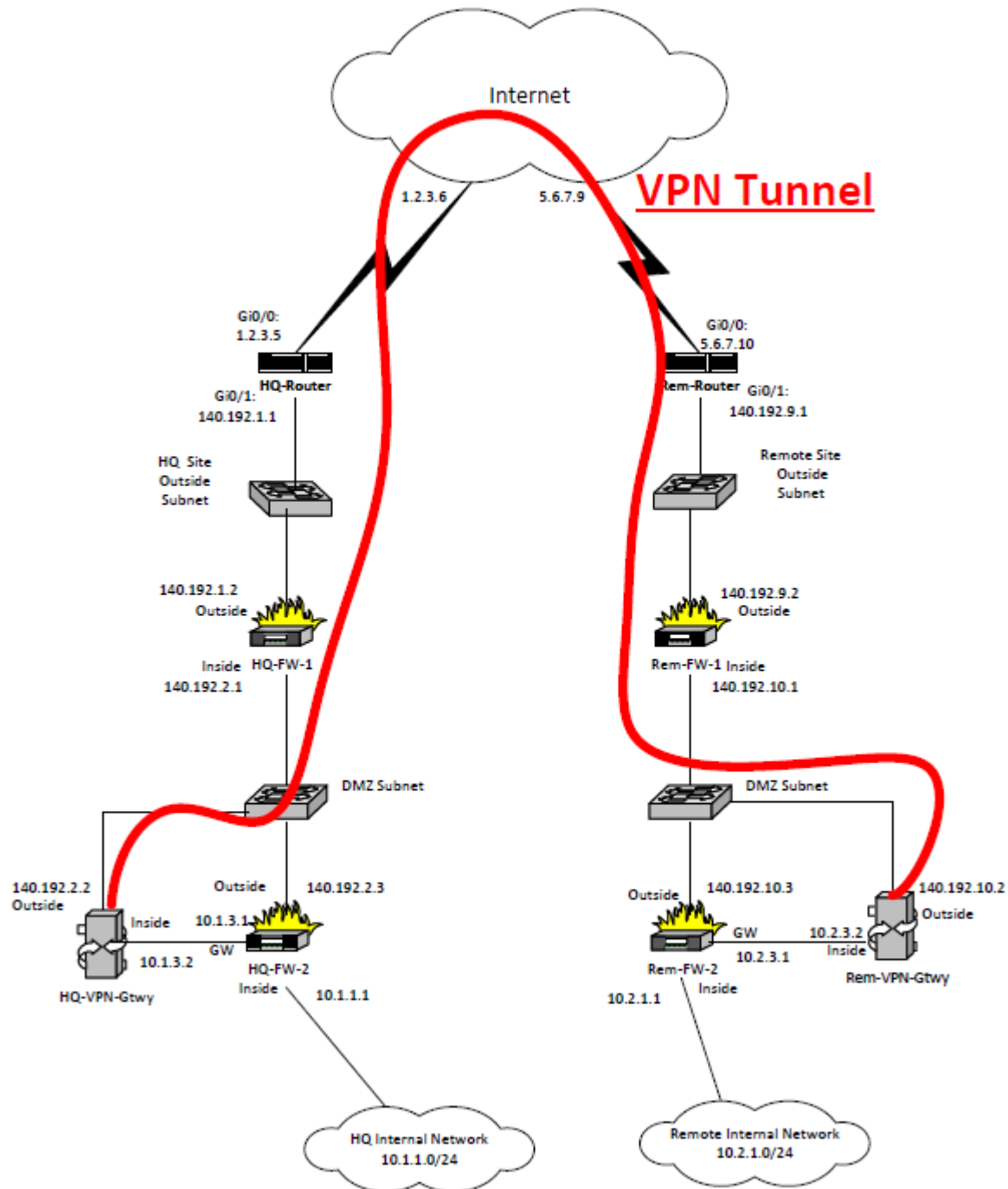


NET 377
Lab 5 – Split-FW Network
Instructions and Answer Document
Due Sunday, March 8, 2026 by 11:59 pm.
50 points

Consider the following split-FW network:



This represents a company network with 2 sites: Headquarters (HQ) and Remote. At each site there is an internal network with private IPs: HQ Internal Network (10.1.1.0/24) and Remote Internal Network (10.2.1.0/24). Each site uses a dual-layer security architecture to connect these internal networks to the public Internet. On each site there are 2 public IP subnets between the internal network and the Internet. A VPN Tunnel is used between the VPN Gateways on each site.

The requirements for packet delivery are:

- Internal network packets between 10.1.1.0/24 and 10.2.1.0/24 will be sent over the VPN Tunnel with no NAT translation. This VPN tunnel is an IPSEC ESP tunnel.
 - Any HQ Internal Network client or Remote Internal Network client can connect to any public Internet server using HTTP or HTTPS over TCP – these connections should use interface PAT.
 - All other packets are denied.
- (8 points) The layer 3 devices (routers, firewalls, VPN gateways) require static routes to be configured so that they can deliver all possible packets to the correct destinations. Open the “Lab5-Static-Routes.xlsx” file and fill in the static route information for all devices in this network diagram. When complete, do a screen capture of this Excel table and paste it here.

Static Route Specifications					
Device Name	Destination Network	Destination Mask	Next Hop IP	Outgoing Interface	Comment
HQ-Router	140.192.2.0	255.255.255.0	140.192.1.2	G0/1	
	0.0.0.0	0.0.0.0	1.2.3.6	G0/0	
HQ-FW-1	0.0.0.0	0.0.0.0	140.192.1.1	G0/0	
	10.1.1.0	255.255.255.0	140.192.2.3	G0/1	
	10.2.1.0	255.255.255.0	140.192.2.2	G0/1	
HQ-FW-2	0.0.0.0	0.0.0.0	140.192.2.1	G0/0	
	10.2.1.0	255.255.255.0	10.1.3.2	G0/1	
HQ-VPN-Gtwy	0.0.0.0	0.0.0.0	140.192.2.3	G0/1	
Rem-Router	0.0.0.0	0.0.0.0	5.6.7.9	G0/0	
	140.192.10.0	255.255.255.0	140.192.9.2	G0/1	
Rem-FW-1	10.2.1.0	255.255.255.0	140.192.10.3	G0/1	
	140.192.0.0	255.255.0.0	140.192.9.1	G0/0	
	0.0.0.0	0.0.0.0	140.192.9.1	G0/0	
Rem-FW-2	0.0.0.0	0.0.0.0	140.192.10.1	G0/0	
	10.1.1.0	255.255.255.0	10.2.3.2	G0/1	
Rem-VPN-Gtwy	10.1.1.0	255.255.255.0	10.2.3.1	G0/1	

- (4 points) Each of the 2 VPN Gateways has a Security Policy Database (SPD) listing information for the VPN Tunnel between them. Open the “Lab5-SPD.xlsx” file and fill in this SPD information. When complete, do a screen capture of this Excel table and paste it here.

Security Policy Database												
IPSEC Device Name	Crypto Map Interface	IPSEC Mode	IPSEC Protocol	SA TRAFFIC MATCH ("Interesting Traffic")						IPSEC Peer IP	SPI (leave blank)	Comments
				Source IP (First)	Source IP (Last or Mask)	Source Port	Dest IP (First)	Dest IP (Last or Mask)	Dest Port			
HQ-VPN-gtwy	G0/0	tunnel	ESP	10.1.1.0	0.0.0.255	any	10.2.1.0	0.0.0.255	any	140.192.10.2		
Rem-VPN-gtwy	G0/0	tunnel	ESP	10.2.1.0	0.0.0.255	any	10.1.1.0	0.0.0.255	any	140.192.2.2		

3. (4 points) The HQ-FW-2 and Rem-FW-2 firewalls need to perform NAT. Open the “Lab5-NAT.xlsx” file and fill in the NAT rules for each of these Firewalls. Note that each of these Firewalls has 3 interfaces named “Inside”, “Outside” and “GW”. When complete, do a screen capture of this Excel table and paste it here.

	Basic NAT Rule Specifications										
Device Name	Inside Interface	Outside Interface	NAT(N) or PAT (P)	Static (S) or Dynamic (D)	Inside IP (First)	Inside IP (Last or Mask)	Outside IP or Interface	Protocol (if specified)	Inside Port Number (if needed)	Outside Port Number (if needed)	Comments
HQ-FW2	G0/1	G0/0	PAT	Dynamic	10.1.1.2	255.255.255.0	140.192.2.3	tcp			
Rem-FW2	G0/1	G0/0	PAT	Dynamic	10.2.1.0	255.255.255.0	140.192.10.3	tcp			

4. (18 points) Create FW rulesets for all 4 firewalls. Open the “Lab5-Rules.xlsx” file and fill in the details for the rulesets. When complete, do a screen capture of this Excel table and paste it here.

Ruleset Specifications										
RuleSet Name	Rule #	P/D	Protocol	Source IP (First)	Source IP (Last or Mask)	Source Port	Dest IP (First)	Dest IP (Last or Mask)	Dest Port	Comments
HQ-FW-2-Inside	2	P	tcp	10.1.1.0	0.0.0.255	any	any	any	80	
	3	P	tcp	10.1.1.0	0.0.0.255	any	any	any	443	
	4	D	any	any	any	any	any	any	any	
HQ-FW-2-Outside	1	P	tcp	any	any	80	140.192.2.3	host	any	
	2	P	tcp	any	any	443	140.192.2.3	host	any	
	3	D	any	any	any	any	any	any	any	
HQ-FW1-Inside	1	P	udp	140.192.10.2	host	any	140.192.2.2	host	500	
	2	P	esp	140.192.10.2	host	any	140.192.2.2	host	any	
	3	P	tcp	any	any	80	140.192.2.3	host	any	
	4	P	tcp	any	any	443	140.192.2.3	host	any	
	5	D	ip	any	any	any	any	any	any	
HQ-FW1-Outside	1	P	tcp	10.1.1.0	0.0.0.255	any	any	any	80	
	2	P	tcp	10.1.1.0	0.0.0.255	any	any	any	443	
	3	P	udp	140.192.2.2	host	any	140.192.10.2	any	500	
	4	P	esp	140.192.2.2	host	any	140.192.10.2	any	any	
	5	D	ip	any	any	any	any	any	any	
Rem-FW1-Inside	1	P	udp	140.192.2.2	host	any	140.192.10.2	host	500	
	2	P	esp	140.192.2.2	host	any	140.192.10.2	host	any	
	3	P	tcp	any	any	80	140.192.10.3	host	any	
	4	P	tcp	any	any	443	140.192.10.3	host	any	
	5	D	ip	any	any	any	any	any	any	
Rem-FW1-Outside	1	P	tcp	10.2.1.0	0.0.0.255	any	any	any	80	
	2	P	tcp	10.2.1.0	0.0.0.255	any	any	any	443	
	3	P	udp	140.192.10.2	host	any	140.192.2.2	host	500	
	4	P	esp	140.192.10.2	host	any	140.192.2.2	host	any	
	5	D	ip	any	any	any	any	any	any	
Rem-FW2-Outside	1	P	tcp	10.2.1.0	0.0.0.255	any	any	any	80	
	2	P	tcp	10.2.1.0	0.0.0.255	any	any	any	443	
	3	D	ip	any	any	any	any	any	any	
Rem-FW2-Inside	1	P	tcp	any	any	80	140.192.10.3	host	any	
	2	P	tcp	any	any	443	140.192.10.3	host	any	
	3	D	ip	any	any	any	any	any	any	

5. (10 points) Implement the lab using CML or Packet Tracer and answer the questions below.
Hint: looks similar to Lab3. The addressing is different. Use routers for VPN-Gateways.
- I didn't implemented the ACLs since the VPN gateway was not working.
6. (3 points) PC A has IP 10.2.1.99 on the Remote Internal Network. PC A sets up an HTTP connection to Internet server B at IP 33.5.6.9. Now PC A sends an HTTP GET Request to B. When this HTTP GET packet (going from A to B) arrives at Rem-Router, what is:
- Source IP address in the packet header:
140.192.10.3
 - Destination IP address in the packet header:
33.5.6.9
 - Provide a screen capture of the packet (using Wireshark or Simulation mode).

PDU Information at Device: Router3

[OSI Model](#) Inbound PDU Details Outbound PDU Details

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 140.192.10.3, Dest. IP: 33.5.6.9

Layer 2: Ethernet II Header
0060.7091.8901 >>
0001.C7B6.D302

Layer 1: Port GigabitEthernet0/1

>

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 140.192.10.3, Dest. IP: 33.5.6.9

Layer 2: Ethernet II Header
0001.C7B6.D301 >>
0060.3E05.1902

Layer 1: Port(s): GigabitEthernet0/0

1. GigabitEthernet0/1 receives the frame.

Challenge Me << Previous Layer Next Layer >>

7. (3 points) PC A has IP 10.2.1.99 at the Remote Internal Network. PC A sets up an HTTP connection to server C, whose IP address is 10.1.1.10, on the HQ Internal network. PC A sends an HTTP GET Request to C. When this HTTP GET packet (going from A to C) arrives at the HQ-Router, what is:
- a) Source IP address in the packet header:
140.192.10.2
 - b) Destination IP address in the packet header:
140.192.2.2
 - c) Provide a screen capture of the packet (using Wireshark or Simulation mode).

PDU Information at Device: Router1

OSI Model

Inbound PDU Details

Outbound PDU Details

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 140.192.10.2, Dest. IP: 140.192.2.2

Layer 2: Ethernet II Header 0060.3E05.1901 >> 00D0.9707.DA01

Layer 1: Port GigabitEthernet0/0

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 140.192.10.2, Dest. IP: 140.192.2.2

Layer 2: Ethernet II Header 00D0.9707.DA02 >> 0050.0F3B.C101

Layer 1: Port(s): GigabitEthernet0/1

1. GigabitEthernet0/0 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>